

# **DIGITAL RADIO LABORATORY**

## **Assignment Report**

*GNU Radio Companion — Waveform Generation, Quantization & FFT Analysis*

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Software Used: GNU Radio Companion (GRC), Python (QT GUI backend)

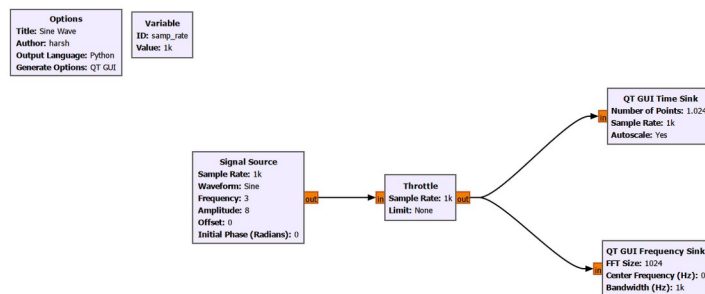
# 1. Introduction

This report documents the design and simulation of a set of signal-processing experiments carried out in GNU Radio Companion (GRC). The experiments cover three broad areas: (i) generation of standard periodic waveforms (sine, cosine, square, triangle and sawtooth) and observation of their time-domain and frequency-domain behaviour, (ii) quantization of a cosine signal using 4-bit and 16-bit quantizers to study the effect of quantization resolution on signal fidelity and noise, and (iii) analysis of the sum of three cosine signals sampled at 8 kHz, used to illustrate the sampling theorem and the phenomenon of aliasing. Each experiment is presented with its GRC flowgraph, the corresponding QT GUI Time Sink / Frequency Sink output, and a short analysis.

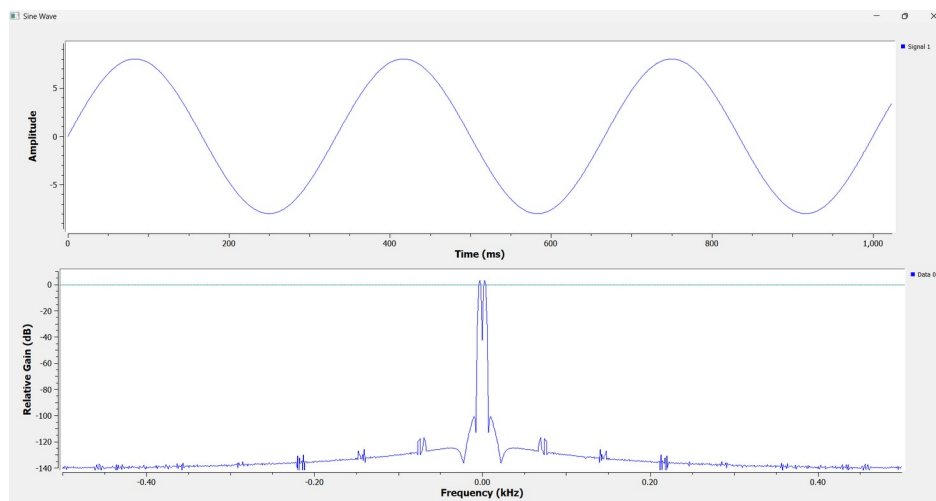
## 2. Generation of Standard Waveforms

For each waveform, a GRC "Signal Source" block was configured with the sample rate  $\text{samp\_rate} = 1 \text{ kHz}$ , frequency = 3 Hz and amplitude = 8 (parameters derived from the roll number as instructed: last three digits give the frequency and the digit-sum gives the amplitude). The Signal Source is followed by a Throttle block (to pace the simulated sample stream to real time) and then split to a QT GUI Time Sink (time-domain view) and a QT GUI Frequency Sink (frequency-domain / FFT view, FFT size = 1024).

### 2.1 Sine Wave



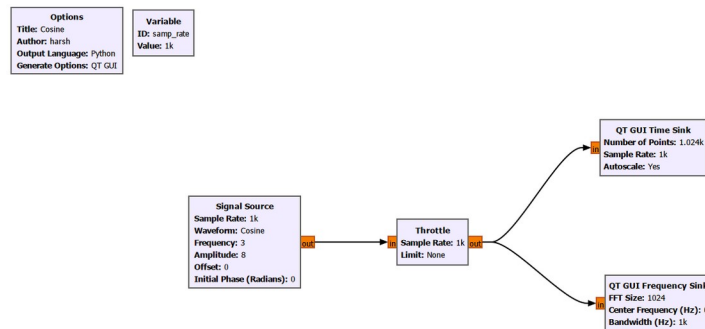
*GRC flowgraph*



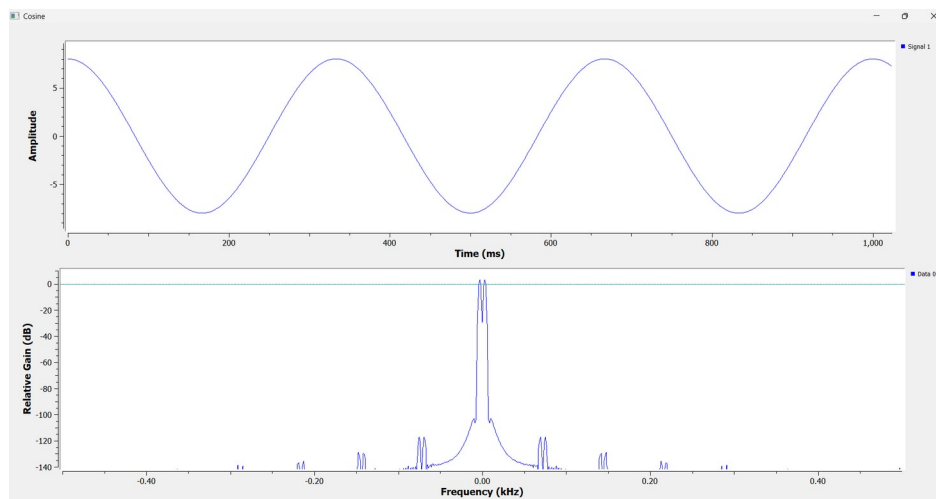
*Time-domain (top) and frequency-domain (bottom) output*

The Signal Source waveform type is set to Sine with Frequency = 3 Hz and Amplitude = 8. The Time Sink shows a smooth sinusoid oscillating between +8 and -8 with a period of about 333 ms ( $\approx 3$  Hz), matching the configured frequency. The Frequency Sink shows a single sharp spectral line at 0 Hz (in the  $\pm 0.5$  kHz span shown, the 3 Hz tone appears essentially at DC on this scale) with the expected symmetric image, confirming that a pure sine wave has all its energy concentrated at a single frequency.

## 2.2 Cosine Wave



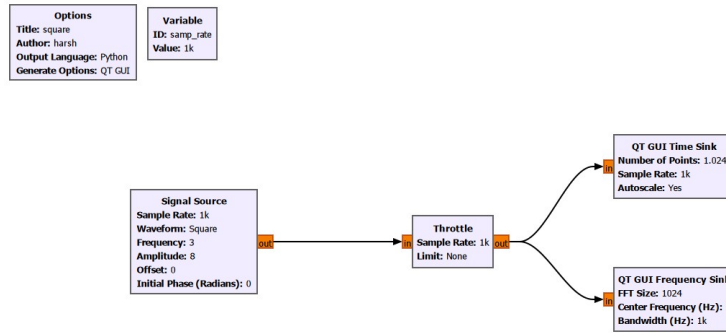
*GRC flowgraph*



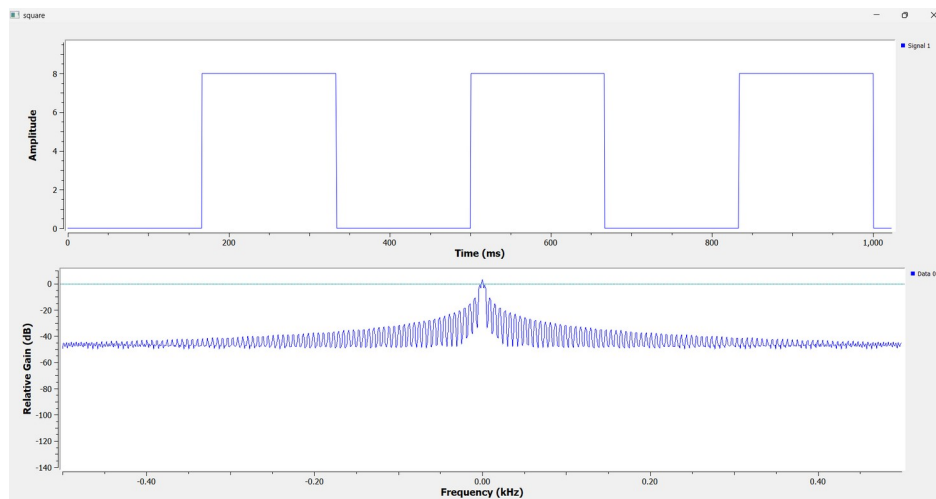
*Time-domain (top) and frequency-domain (bottom) output*

The waveform type is changed to Cosine, keeping Frequency = 3 Hz and Amplitude = 8. The time-domain trace is identical in shape to the sine wave but shifted in phase — it starts at its maximum value (+8) at  $t = 0$  instead of at zero, which is the defining  $90^\circ$  phase difference between sine and cosine. The frequency response again shows a single-tone spectrum with a sharp peak, identical in shape and location to the sine wave's spectrum, since a phase shift does not change the magnitude spectrum of a signal.

## 2.3 Square Wave



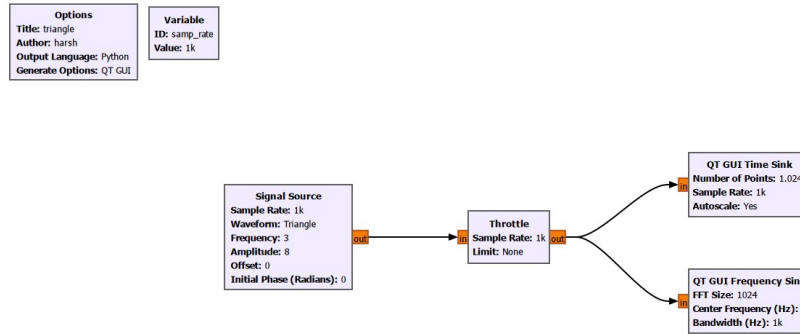
*GRC flowgraph*



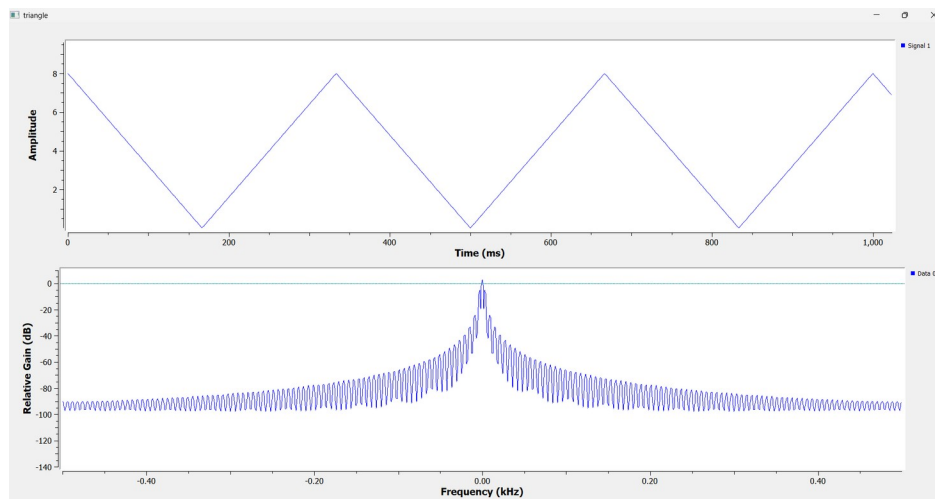
*Time-domain (top) and frequency-domain (bottom) output*

The waveform type is changed to Square, with Frequency = 3 Hz and Amplitude = 8. The Time Sink shows the expected two-level (0 to 8) periodic square pulse train. Because a square wave can be represented as an infinite Fourier series of odd harmonics (fundamental + 3rd + 5th + 7th ... with decreasing amplitude), the Frequency Sink shows a dominant tone at the fundamental frequency accompanied by a comb of smaller harmonic components spaced at odd multiples of the fundamental, rather than a single clean spectral line.

## 2.4 Triangle Wave



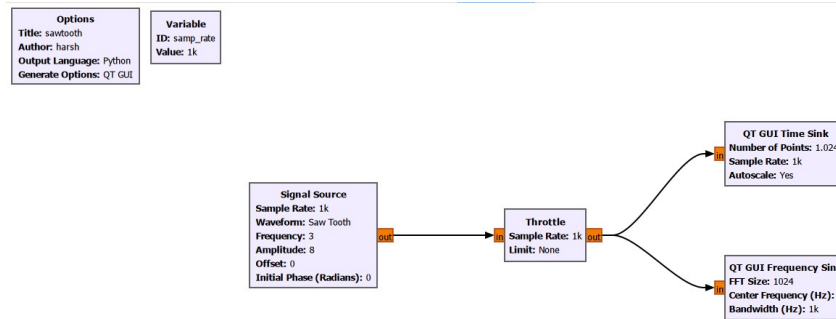
*GRC flowgraph*



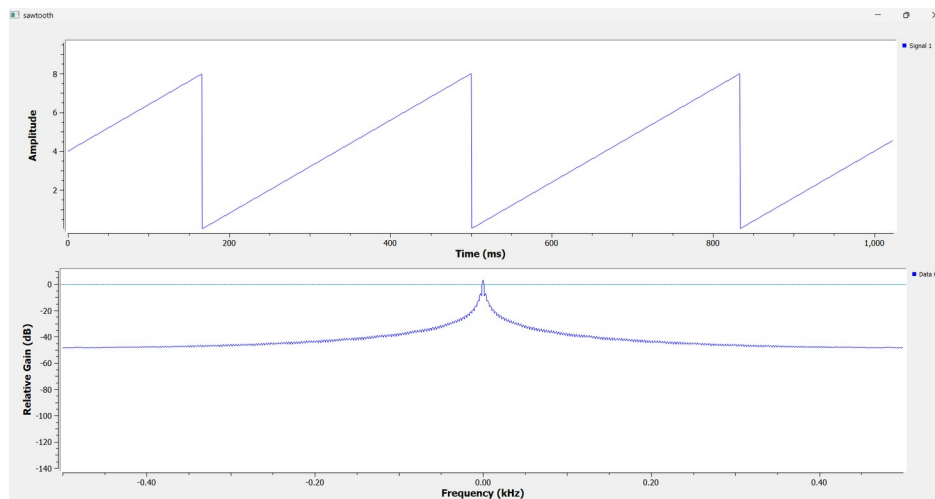
*Time-domain (top) and frequency-domain (bottom) output*

The waveform type is changed to Triangle, with the same Frequency = 3 Hz and Amplitude = 8. The time-domain plot shows the characteristic linear rise-and-fall (triangular) shape between 0 and 8. Like the square wave, the triangle wave's Fourier series contains only odd harmonics, but because their amplitudes fall off much faster (as  $1/n^2$  compared to  $1/n$  for the square wave), the harmonic sidelobes visible in the Frequency Sink decay much more quickly than in the square-wave case.

## 2.5 Sawtooth Wave



*GRC flowgraph*



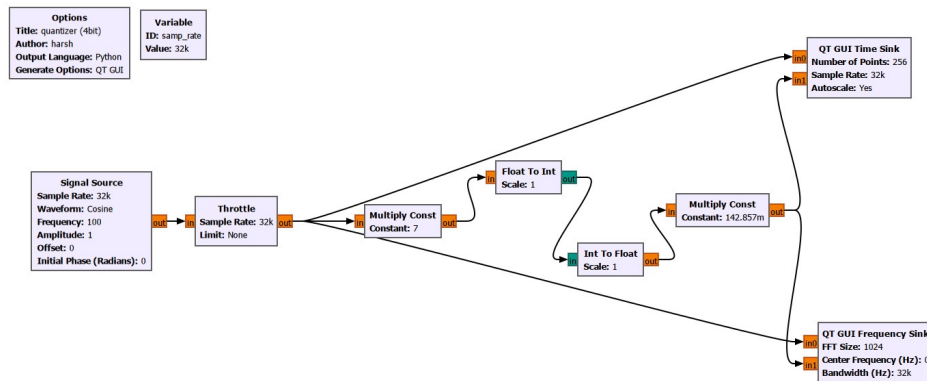
*Time-domain (top) and frequency-domain (bottom) output*

The waveform type is changed to Saw Tooth, with Frequency = 3 Hz and Amplitude = 8. The Time Sink shows the ramp-up / sharp-drop pattern characteristic of a sawtooth. Its Fourier series contains both even and odd harmonics (amplitude falling as  $1/n$ ), so the Frequency Sink shows a fundamental peak together with a full comb of both even- and odd-order harmonics, richer than the triangle wave's spectrum but with faster harmonic roll-off than the square wave.

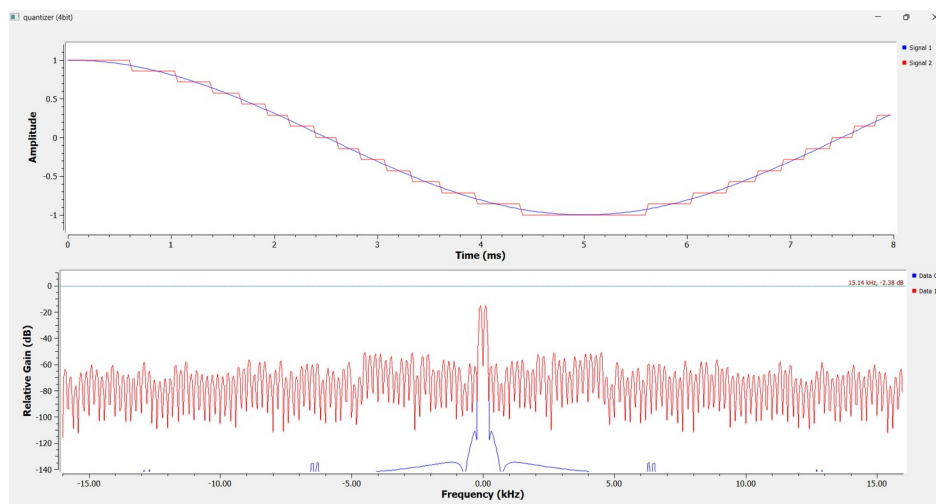
### 3. Quantization of a Cosine Signal

A cosine signal (Sample Rate = 32 kHz, Frequency = 100 Hz, Amplitude = 1) is passed through a Throttle block and then quantized using a chain of Multiply-Const → Float-to-Int → Int-to-Float → Multiply-Const blocks. This chain implements a uniform quantizer: the input is scaled up by  $2^{n-1}$  ( $n$  = number of bits), rounded to the nearest integer (Float to Int), converted back to floating point (Int to Float), and scaled back down by the reciprocal factor, so that the output signal only takes a finite number of discrete amplitude levels (quantization steps) instead of a continuous range of values. Both the original cosine (Signal 1 / blue) and the quantized version (Signal 2 / red) are displayed together on the QT GUI Time Sink and Frequency Sink so the quantization error / added noise can be compared directly.

#### 3.1 4-bit Quantizer ( $n = 4$ )



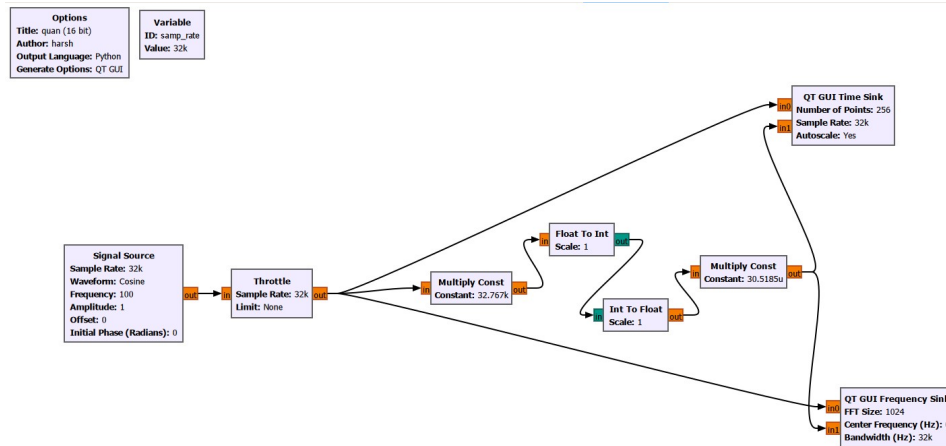
GRC flowgraph — 4-bit quantizer (scale constant = 7, i.e.  $2^3 - 1$ )



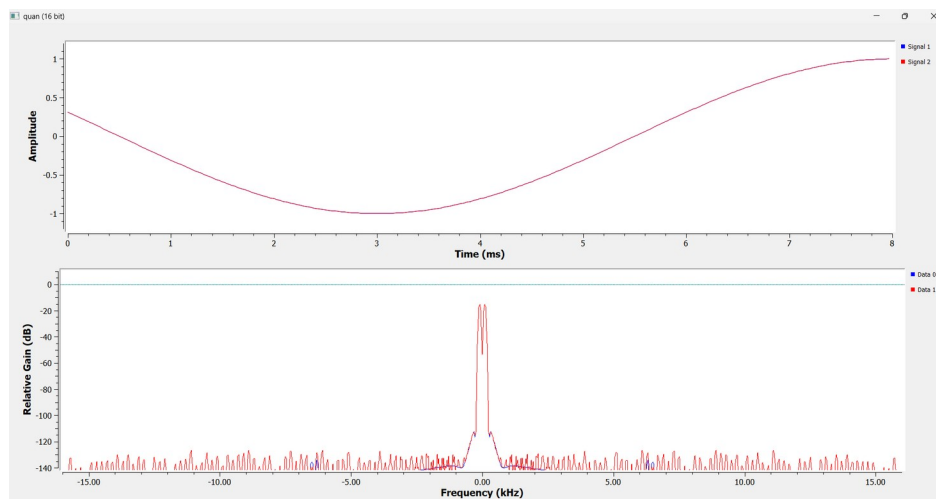
Time-domain (top) and frequency-domain (bottom) output — original cosine vs. 4-bit quantized cosine

With only  $n = 4$  bits, the quantizer has just  $2^4 = 16$  discrete amplitude levels available across the  $\pm 1$  range. This coarse step size is clearly visible in the time-domain plot: the red (quantized) trace follows the blue cosine only approximately, forming a clearly visible staircase pattern with large, easily-distinguishable steps. In the frequency domain, this large quantization error manifests as a raised noise floor across the whole spectrum (quantization noise), and the fundamental tone sits only about  $-2$  dB relative to the reference level with the noise floor sitting much closer to the signal peak than in the 16-bit case, indicating a poor signal-to-quantization-noise ratio (SQNR).

### 3.2 16-bit Quantizer (n = 16)



GRC flowgraph — 16-bit quantizer (scale constant  $\approx 32.767k$ , i.e.  $2^{15} - 1$ )



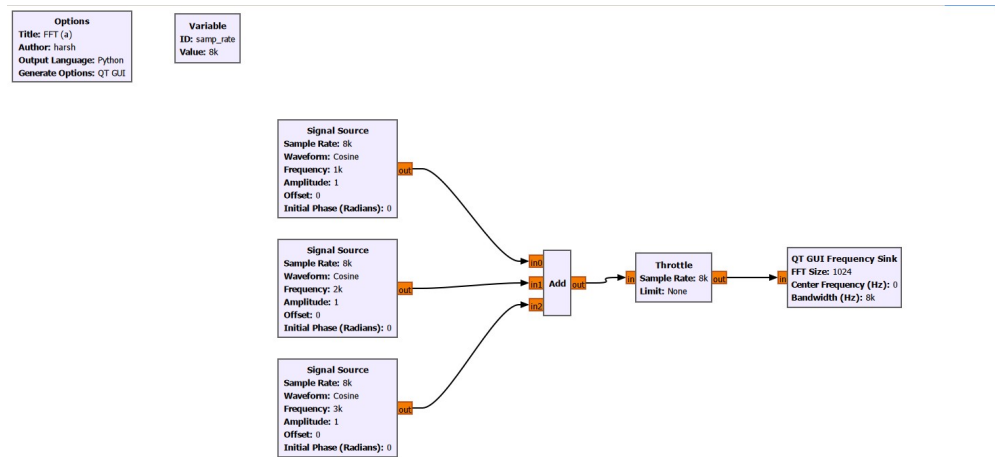
Time-domain (top) and frequency-domain (bottom) output — original cosine vs. 16-bit quantized cosine

With  $n = 16$  bits the quantizer now has  $2^{16} = 65536$  discrete levels, so the quantization step size becomes extremely small. In the time-domain plot the red (quantized) and blue (original) traces are visually indistinguishable — they overlap almost perfectly into a single smooth cosine curve, since the staircase steps are far too fine to be seen at this scale. In the frequency domain, the quantized signal's spectrum sits almost exactly on top of the original signal's spectrum, and the quantization-noise floor is pushed down dramatically compared to the 4-bit case. This confirms the well-known result that SQNR improves by roughly 6 dB for every additional quantization bit, so a 16-bit quantizer (12 extra bits over the 4-bit case) gives on the order of 70 dB better SQNR.

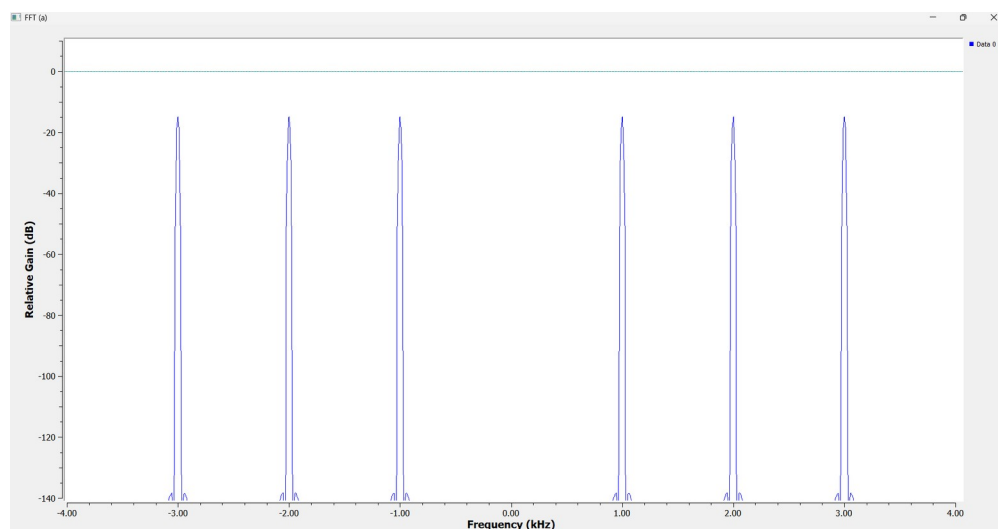
## 4. FFT of Three Summed Cosine Signals ( $f_s = 8$ kHz)

Three cosine Signal Source blocks (each amplitude = 1, sample rate  $f_s = 8$  kHz) are added together using an Add block, throttled to real time, and observed on a QT GUI Frequency Sink (FFT size = 1024, span =  $\pm 4$  kHz since  $f_s = 8$  kHz).

### 4.1 Case (a): $f_1 = 1$ kHz, $f_2 = 2$ kHz, $f_3 = 3$ kHz



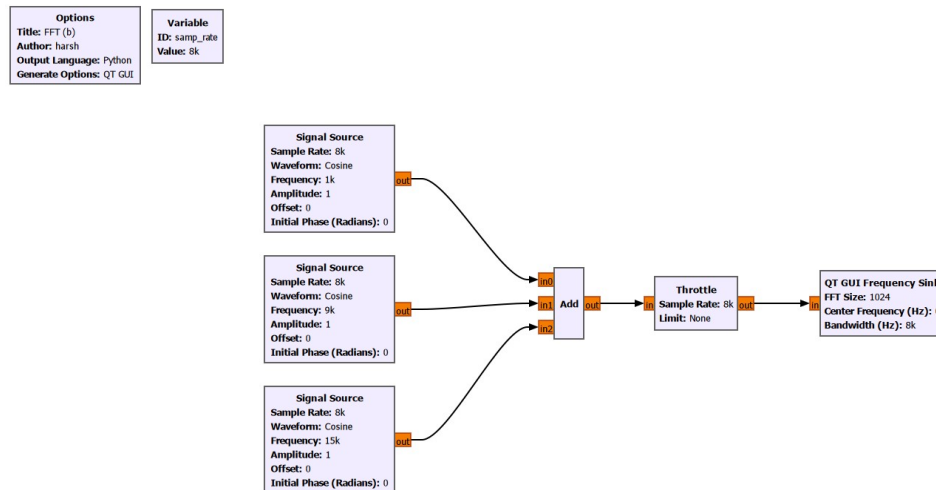
GRC flowgraph — three cosines (1k, 2k, 3k) summed,  $f_s = 8$  kHz



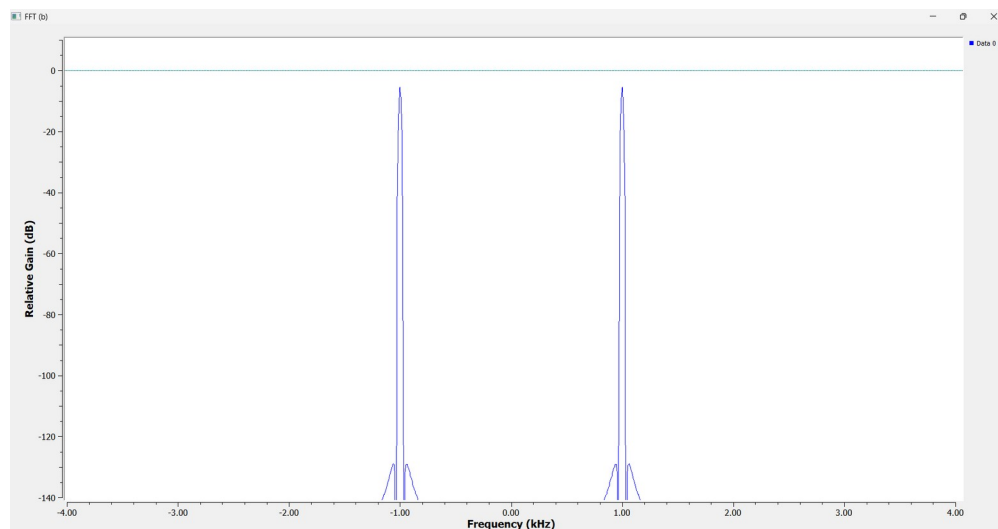
FFT output — case (a)

All three tones (1 kHz, 2 kHz, 3 kHz) lie well below the Nyquist frequency of  $f_s/2 = 4$  kHz, so no aliasing occurs. The Frequency Sink shows exactly six clean spectral peaks: three positive-frequency peaks at +1 kHz, +2 kHz and +3 kHz (each representing one of the input cosines) and their mirror-image peaks at -1 kHz, -2 kHz and -3 kHz (since the FFT of a real signal is always conjugate-symmetric about 0 Hz). Each peak appears exactly where expected, correctly reproducing the frequency content of the input.

## 4.2 Case (b): $f_1 = 1$ kHz, $f_2 = 9$ kHz, $f_3 = 15$ kHz



GRC flowgraph — three cosines (1k, 9k, 15k) summed,  $f_s = 8$  kHz



FFT output — case (b)

Here  $f_2 = 9$  kHz and  $f_3 = 15$  kHz both exceed the Nyquist frequency of 4 kHz (half of  $f_s = 8$  kHz), so they are undersampled and appear as aliases folded back into the  $\pm 4$  kHz base band. The alias of a frequency  $f$ , when sampled at  $f_s$ , appears at  $|f - k \cdot f_s|$  for the integer  $k$  that brings the result into the Nyquist range. For  $f_2 = 9$  kHz:  $9 \text{ kHz} - 8 \text{ kHz} = 1 \text{ kHz}$ , so it aliases down to 1 kHz. For  $f_3 = 15$  kHz:  $15 \text{ kHz} - 2 \times 8 \text{ kHz} = -1 \text{ kHz} \rightarrow |-1 \text{ kHz}| = 1 \text{ kHz}$ , so it also aliases to 1 kHz. As a result, all three input tones (1 kHz, 9 kHz and 15 kHz) collapse onto the very same pair of spectral lines at  $\pm 1$  kHz in the observed spectrum — instead of six distinct peaks as in case (a), only two peaks (at +1 kHz and -1 kHz) are visible, exactly as seen in the plot.

## 4.3 Conclusions from (a) and (b)

- When every input frequency is below the Nyquist frequency ( $f_s/2$ ), the FFT correctly reproduces each individual tone as a distinct spectral peak — case (a) shows six well-separated peaks for the three cosines and their mirror images.
- When one or more input frequencies exceed  $f_s/2$ , they are indistinguishable from a lower "alias" frequency after sampling — case (b) shows that 1 kHz, 9 kHz and 15 kHz all sample down to the same apparent 1 kHz component, so the reconstructed spectrum is misleading and information about the true signal content is permanently lost.

- This directly demonstrates why the sampling rate must be chosen according to the sampling theorem ( $f_s > 2 \times f_{\max}$ ) — otherwise higher-frequency components masquerade as false lower-frequency components in the sampled data, and this cannot be undone after the fact.

## 4.4 Sampling Theory

The Nyquist–Shannon sampling theorem states that a continuous-time, band-limited signal  $x(t)$  whose highest frequency component is  $f_{\max}$  can be completely and unambiguously represented by, and reconstructed from, a sequence of its instantaneous samples — provided the sampling frequency  $f_s$  satisfies:

$$f_s > 2 \times f_{\max}$$

The quantity  $2 \times f_{\max}$  is called the Nyquist rate, and  $f_s/2$  is called the Nyquist frequency. When this condition is satisfied, the original continuous-time signal can theoretically be perfectly reconstructed from its samples using an ideal low-pass (reconstruction/interpolation) filter with cutoff at  $f_s/2$ . Sampling in the time domain corresponds, in the frequency domain, to periodic replication of the signal's spectrum at intervals of  $f_s$ . If  $f_s > 2f_{\max}$ , these spectral replicas (images) are spaced far enough apart that they do not overlap, so the original baseband spectrum can be recovered cleanly by low-pass filtering. This theorem underlies all analog-to-digital conversion — it dictates the minimum sampling rate needed to digitize a signal (for example, audio CDs use  $f_s = 44.1$  kHz because the audible range extends to about 20 kHz, giving comfortable margin above the 40 kHz Nyquist rate).

## 4.5 Aliasing Error and Its Prevention

Aliasing is the distortion that occurs when a signal is sampled at a rate lower than the Nyquist rate ( $f_s \leq 2f_{\max}$ , i.e. the signal contains frequency components at or above  $f_s/2$ ). Under these conditions, the periodic spectral replicas produced by sampling overlap with one another. A high-frequency component  $f > f_s/2$  then becomes indistinguishable from — and is misread as — a lower "alias" frequency given by  $f_{\text{alias}} = |f - k \cdot f_s|$  for the integer  $k$  that folds  $f$  into the range  $[0, f_s/2]$ . Once this folding happens, the true high-frequency content is irrecoverably lost and corrupts the reconstructed signal with false low-frequency content; case (b) of this report (Section 4.2) is a direct demonstration, where 9 kHz and 15 kHz tones both alias down to 1 kHz when sampled at  $f_s = 8$  kHz.

Aliasing can be prevented by:

- Anti-aliasing (low-pass) filtering: passing the analog signal through a low-pass filter with cutoff at  $f_s/2$  before sampling, so that any energy above the Nyquist frequency is removed and cannot fold back into the baseband.
- Increasing the sampling rate: choosing  $f_s$  comfortably greater than  $2 \times f_{\max}$  (oversampling) so that the Nyquist frequency exceeds the highest frequency actually present in the signal.
- Band-limiting the source signal at the design stage, wherever possible, so that  $f_{\max}$  is known and controlled and an appropriate  $f_s$  can be selected in advance.

## 5. Overall Conclusion

This set of experiments illustrated, using GNU Radio Companion, how the shape of a periodic waveform (sine, cosine, square, triangle, sawtooth) determines the richness of its harmonic content in the frequency domain; how the number of quantization bits directly controls the granularity of a digitized signal and its quantization-noise floor (4-bit quantization introduces clearly visible distortion and a high noise floor, while 16-bit quantization is virtually indistinguishable from the original analog signal); and how respecting or violating the Nyquist sampling criterion determines whether an FFT correctly represents a signal's true frequency content or produces misleading aliased components. Together these results reinforce the practical importance of the sampling theorem and anti-aliasing filtering in any real-world analog-to-digital signal processing chain.